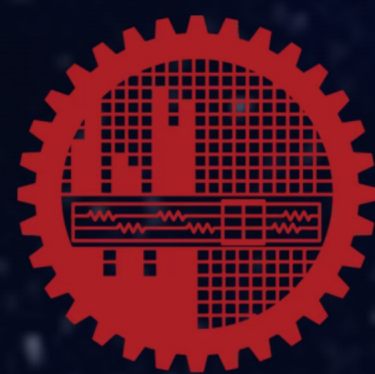




# Automated High-Magnification Alt-Azimuth Telescope

Stellarium-guided, ESP32-driven dual-axis refractor with 120:1 worm reduction

## Abstract

An automated Alt-Az refractor (D80 F900) driven by a three-tier control chain: a Python hub with Stellarium, an ESP32 pulse generator, and microstepped NEMA 17 motors on custom 120:1 worm drives, plus camera exposure control.

## System Architecture

- 1** Stellarium & Python Hub  
Targeting and RA/Dec to Alt/Az conversion
- 2** ESP32 Controller  
Trapezoidal profiling; deterministic step pulses
- 3** Motor Drivers  
1:16 microstepping, phase-current conditioning
- 4** NEMA 17 + 120:1 Worm Drive  
Dual-axis actuation with anti-backdrive holding

**Camera module:** shutter-timing exposure control and an intervalometer for sequential capture and stacking.



## Drive Kinematics

**120:1**  
WORM RATIO

**1:16**  
MICROSTEP

**384,000**  
STEPS / REV

**3.375"**  
PER MICROSTEP

**f/11.25**  
D80 F900

**72×**  
MAGNIFICATION

$$200 \times 16 \times 120 = 384,000 \text{ steps per axis revolution}$$

$$360^\circ \div 384,000 \approx 3.375'' \text{ per step}$$

Theoretical commanded resolution; real accuracy depends on printed-gear backlash and flex.

## Engineering Challenges

- **Machining:** metal 120:1 worm set failed on tooling and pitch tolerances; moved to 3D printing.
- **Shearing:** printed worm screw failed under torque; flexure caused binding and fatigue fracture.
- **Redesign:** pitch, infill, print orientation and sleeve bearings revised over several cycles.
- **Control:** Python ephemeris updates synchronised with ESP32 pulses, no lost steps.

## Technology Stack

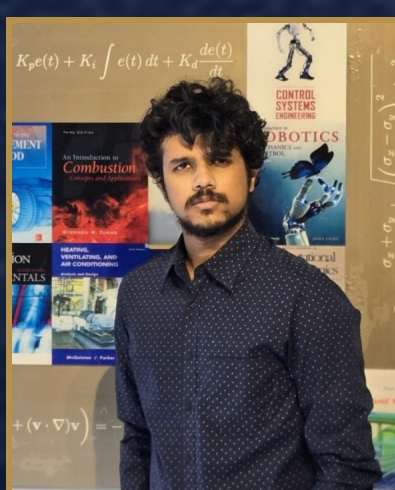
|            |                                        |
|------------|----------------------------------------|
| SOFTWARE   | Python hub with Stellarium integration |
| CONTROLLER | ESP32 dual-core SoC                    |
| MOTORS     | NEMA 17, 1.8° step, 1:16 microstepping |
| DRIVE      | 3D-printed worm and 120-tooth wheel    |
| OPTICS     | D80 F900 achromatic refractor          |

## Future Enhancements

- Rigid, vibration-damped camera mount for lunar imaging.
- Right-angle vertical focuser for near-zenith viewing.
- Cloud-based control server for mobile operation.

## Project Team

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## Open Repository

[github.com/M000nwalker/Beyond\\_Horizon\\_1](https://github.com/M000nwalker/Beyond_Horizon_1)

Design files & source code